

Chapter 4 : part 1

① Sum Rule / Law of Total Probability

$$(a) P_X(x) = \sum_y P_{X|Y}(x|y) \times P_Y(y)$$

$$(b) P_X(x) = \sum_y P_{X,Y}(x,y)$$

② Product Rule

$$P_{X,Y}(x,y) = P_{Y|X}(y|x) P_X(x)$$

③ Conditional Expectation

$$\text{if } E[X] = \sum_x x P_X(x),$$

$$\text{then } E[X|A] = \sum_x x P_{X|A}(x|a)$$

④ Law of Total ~~Expectation~~ Expectation

$$(a) E[X] = \sum_y E[X|Y=y] P_Y(y)$$

$$(b) E[Y] = \sum_x E[Y|X=x] P_X(x)$$

Bayes Theorem for Discrete RV

$$P_{X|Y}(x|y) = \frac{P_{Y|X}(y|x) P_X(x)}{\sum_x P_{Y|X}(y|x) P_X(x)}$$

$$\Rightarrow P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$$

Conditioned on Another RV

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$